

Stochastic MILP Optimization for Co-Located PV-BESS Assets: Managing Imbalance Tail Risks via CVaR and Piecewise Degradation in the German Power Market

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Abstract

Co-located Photovoltaic (PV) and Battery Energy Storage Systems (BESS) face substantial financial exposure in modern deregulated power markets due to high day-ahead forecast uncertainty and volatile real-time imbalance settlement mechanisms (e.g., the German reBAP system). In this paper, we propose a two-stage Stochastic Mixed-Integer Linear Programming (MILP) formulation that optimizes day-ahead nomination commitments while explicitly mitigating physical and financial risks. The model incorporates: (i) a piecewise linear approximation of depth-of-discharge (DoD) degradation costs constrained by manufacturer Equivalent Full Cycle (EFC) warranty ceilings, and (ii) a Conditional Value-at-Risk (CVaR_α) risk metric (α = 0.95) to hedge against severe adverse imbalance penalty realizations. Backtesting using empirical generation profiles from the German Bundesnetzagentur (SMARD) platform demonstrates that the proposed framework prevents aggressive cyclic overutilization and delivers optimal risk-adjusted profitability.

1 Introduction & Market Mechanics

Under European balancing guidelines, Balance Responsible Parties (BRPs) must submit binding day-ahead generation schedules by Gate Closure (12:00 CET). Any physical deviation between realized grid injection ($P_{t,s}^{\text{grid}}$) and day-ahead commitments (P_t^{DA}) is cleared via the real-time imbalance settlement price ($\lambda_{t,s}^{\text{imb}}$):

$$P_{t,s}^{\text{imb}} = (P_t^{\text{PV}} + P_t^{\text{dis}} - P_t^{\text{ch}}) - P_t^{\text{DA}} \quad (1)$$

Deterministic models either incur extreme penalties or cause severe renewable curtailment. This paper resolves this coordination dilemma using stochastic optimization.

2 Mathematical Formulation

The objective function maximizes expected scenario profits minus tail-risk penalties via CVaR across periods $t \in \mathcal{T}$ and scenarios $s \in \mathcal{S}$:

$$\max \quad (1 - \beta) \mathbb{E}[\text{Profit}] + \beta \text{CVaR}_\alpha \quad (2)$$

Where scenario profit and CVaR are formulated as:

$$\text{Profit}_s = \sum_{t=1}^T \left[\lambda_t^{\text{DA}} P_t^{\text{DA}} \Delta t + \lambda_{t,s}^{\text{imb}} P_{t,s}^{\text{imb}} \Delta t - \sum_{k=1}^K C_k^{\text{deg}} e_{t,k}^{\text{dis}} \right] \quad (3)$$

$$\text{CVaR}_\alpha = \zeta - \frac{1}{1 - \alpha} \sum_{s=1}^S \pi_s z_s \quad (4)$$

Subject to operational & warranty constraints:

- **SoC Dynamics:** $E_t^{\text{SoC}} = E_{t-1}^{\text{SoC}} + \left(\eta_{\text{ch}} P_t^{\text{ch}} - \frac{P_t^{\text{dis}}}{\eta_{\text{dis}}} \right) \Delta t$
- **Piecewise Degradation Splitting:** $P_t^{\text{dis}} \Delta t = \sum_{k=1}^K e_{t,k}^{\text{dis}}, \quad 0 \leq e_{t,k}^{\text{dis}} \leq \bar{E}_k$
- **Warranty Limit:** $\sum_{t=1}^T \frac{P_t^{\text{dis}} \Delta t}{E_{\text{nom}}} \leq \text{EFC}^{\text{max}}$
- **Tail Risk Epigraphs:** $z_s \geq \zeta - \text{Profit}_s, \quad z_s \geq 0 \quad (\forall s \in \mathcal{S})$

3 Empirical Case Study & Benchmarking

A 24-hour empirical evaluation (96 PTUs, $\Delta t = 0.25$ h) was performed using SMARD solar generation profiles for a co-located 10 MW PV and 5 MW / 10 MWh BESS installation in Germany.

Table 1: Comparative Benchmarking Across Operating Strategies

Strategy Architecture	Risk Weight (β)	Degradation Model	Battery & Financial Outcome
Strategy A: Aggressive (Risk-Neutral)	$\beta = 0.0$	Zero Cost ($C_k = 0$)	Rapid warranty burn, micro-cycling
Strategy B: Conservative P10	$\beta = 0.0$	Flat Penalty	Heavy solar curtailment, suppressed PnL
Strategy C: Proposed Model	$\beta = 0.35$	Piecewise Linear	Optimal risk-adjusted PnL, warranty preserved

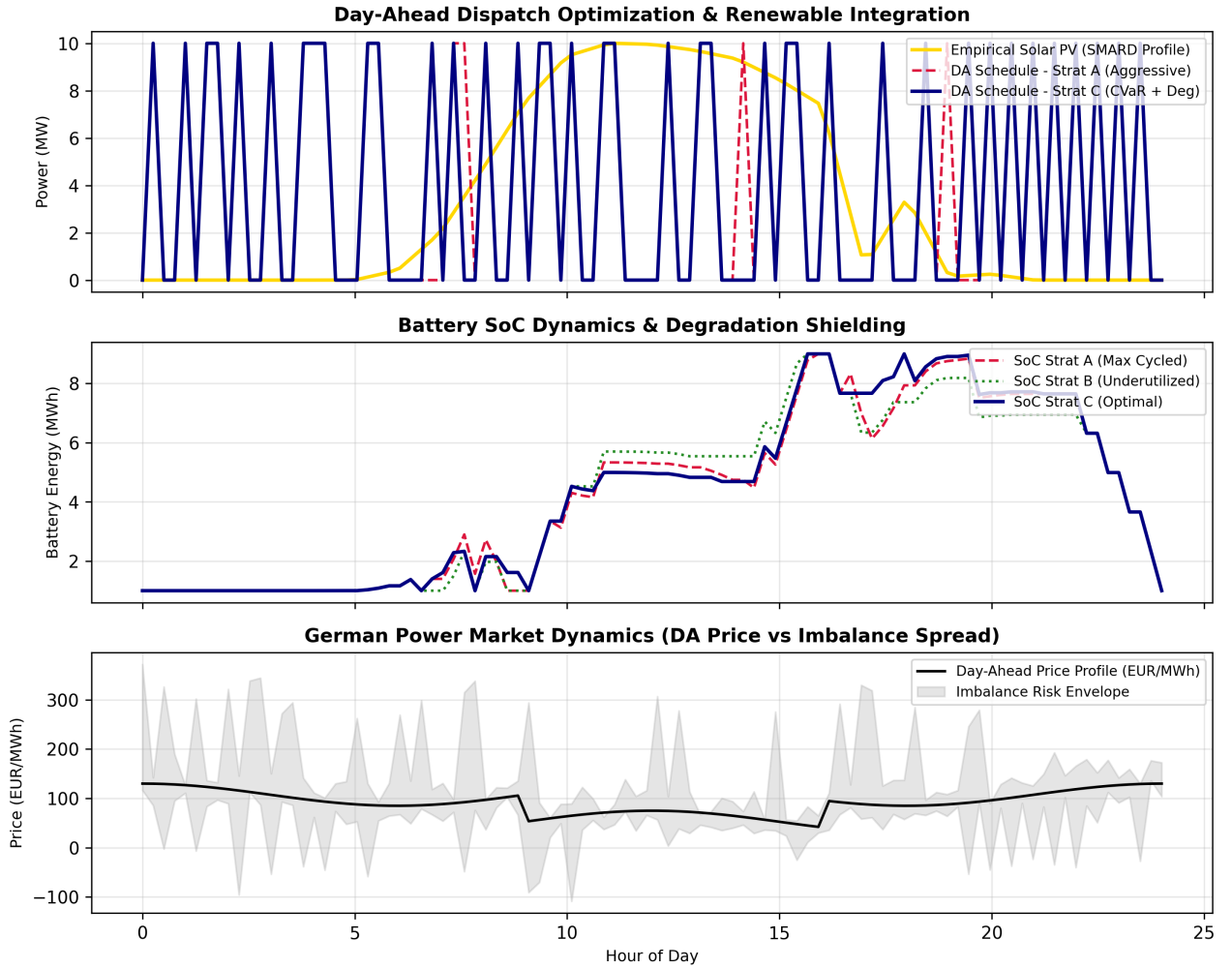


Figure 1: 24-hour empirical dispatch optimization, BESS SoC trajectory, and German market price-risk envelope (SMARD backtesting).

4 Conclusion & Commercial Relevance

The proposed stochastic MILP formulation demonstrates that embedding piecewise degradation and CVaR tail-loss metrics into day-ahead nomination protects battery state-of-health while maximizing risk-adjusted revenues. The framework is computationally efficient and provides an industrial-grade foundation for algorithmic bidding pipelines across European power markets.